(b) How you will determine the uplift pressure on the roof of a syphon aqueduct? 7

(c) Discuss the possible causes of failure of cross drainage work. 7

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(d) What is a super-passege? Draw a neat sketch of super-passege? 7

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B. E. (Eighth Semester) Examination, April-May, 2016

(Old Scheme)

(Civil Engg. Branch)

WATER RESOURCES ENGINEERING–II

Time Allowed : Four hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Part (a) of each question is compulsory. Attempt any two part from (b), (c) and (d). Graph paper may be provided to students on dimand.*

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1. (a) What is spillway? 2

(b) Briefly describe different types of dams with their advantages and disadvantages. 7

(c) Write short notes on :

(i) Arch Dam

(ii) Rock fill dam

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(d) Discuss the details factors that must be considered when specifying a particular type of dam for particular project. 7

2. (a) What is a gravity dam? 2

(b) Sketch an ogee spillway profile and made in it the different zones. 7

(c) Write a short notes on the energy dissipation below spillways. 7

(d) Determine the number of siplion spillway units required with the following data :

High flood level = 375·50

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Full Reserviour level = 374·80

Level of centre of siphon outlet = 368·30

High flood discharge = 580 cumecs

Width of siphon throat = 4·5 m

Height of siphon throat = 2·0 m

The siphon of spillway discharges freely in the air.

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3. (a) Differentiate Weir and a barrage. 2

(b) Sketch a 'Typical Layout of Diversion Head Work' and explain briefly. 7

(c) What is Bligh's creep theory? How does Khosla's theory differ from TL? 7

(d) With the help of a neat sketch, describe a divide wall. 7

4. (a) What are the purposes of regulation work? 2

(b) With the help of neat sketch, describe a distributory head regulator. 7

(c) How classification of fall is done? Describe. 7

(d) What is meant by canal escape? With the help of a neat sketch describe an escape 7

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5. (a) What is a level crossing? 2